

Absolute Value Functions

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $(-\infty, -6] \cup [-2, \infty)$

_____ State the vertex of $f(x) = |x + 1| - 3$.

ANSWER: $(-6, 6)$

_____ Solve $|x - 3| \leq 5$. (interval notation)

ANSWER: no solution

_____ Solve $3|x| - 5 = 10$. (write the solution set)

ANSWER: $\{-5, 13\}$

_____ Solve $|2x + 1| = 7$. (write the solution set)

ANSWER: $(-\infty, -4) \cup (3, \infty)$ # _____ Solve $x + 4 \geq 2$. (interval notation)	ANSWER: $\{-5, 5\}$ # _____ Solve $x < 6$. (interval notation)
ANSWER: $(-1, -3)$ # _____ State the vertex of $f(x) = 2 x - 3 + 4$.	ANSWER: $(3, 4)$ # _____ Solve $x - 4 = 9$. (write the solution set)
ANSWER: $[-2, 8]$ # _____ Solve $2x + 1 > 7$. (interval notation)	ANSWER: $\{-4, 3\}$ # _____ Solve $x + 6 + 8 = 3$.